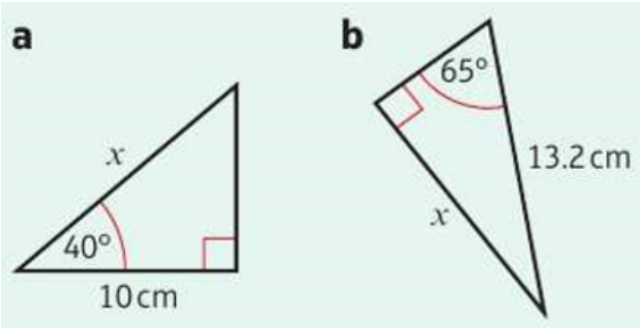
# Moments

## Solutions

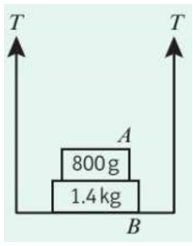
### Starter

Solution 2

1. Find $x$ in each right-angled triangle.




2. The light scale-pan is supported by two strings, each of tension $T$. Find $T$, the reaction of $B$ on $A$, and the reaction of the pan on $B$.




1a–b

a) $x = 10/\cos 40° = 13.1$ cm.

b) $x = 13.2 \sin 65° = 12.0$ cm.

2

$2T = 2.2g$, so $T = 1.1g \approx 11$ N.

$R_{B \text{ on } A} = 0.8g \approx 7.8$ N.

$R_{\text{pan on } B} = 2.2g \approx 22$ N (2 s.f.).

## The door problem

Solution 3

Why is a door handle usually far from the hinge?




Turning effect

For a given force, increasing its perpendicular distance from the hinge increases the turning effect.

If the perpendicular distance is doubled, what happens to the force required to produce the same turning effect?

Double the distance

The required force halves. The product of force and perpendicular distance stays the same.

## Rigid Bodies and Overview

Solution 4

### From particles to rigid bodies

A particle model ignores an object's dimensions. A rigid-body model includes its size and shape, so it can describe rotation.

A rod is a thin rigid body. A lamina is a thin, flat body.

### Three ideas in this chapter

1. Equilibrium: the resultant force and resultant moment are both zero.
2. Centre of mass: a uniform rod's weight acts at its midpoint; a non-uniform rod's need not.
3. Tilting: a support about to lose contact exerts no reaction. Use the remaining pivot to take moments.

## Moments

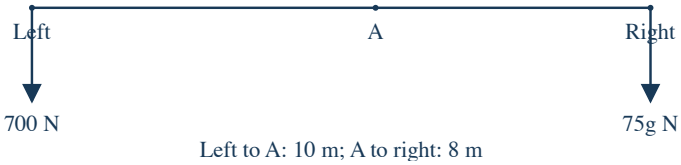
Solution 5

**Moment of a force about a point**

$M = Fd$

Here $d$ is the **perpendicular distance** from the point to the force's line of action.

Units: newton metres, N m.




A light seesaw carries a 700 N weight on the left and a 75 kg mass on the right. Which way does it turn about $A$?

Moments

Anticlockwise: $700(10) = 7000$ N m.

Clockwise: $75g(8) = 5880$ N m.

Direction

The anticlockwise moment is greater: resultant 1120 N m anticlockwise.

## Quickfire Examples

Solution 6

Find each moment about $P$, including its direction.

1. A 20 N force acts downwards 10 m to the right of $P$.
2. A 20 N force has a line of action through $P$.
3. A 10 N force acts at 60° below a horizontal rod, 5 m to the right of $P$.
4. A 10 N force acts at 30° above a horizontal rod, 2 m to the right of $P$.
5. A 5 N force acts 65° from the upward vertical, 3 m to the right of $P$.

1–3

1. $20(10) = 200$ N m clockwise.
2. Zero: perpendicular distance is zero.
3. $10(5\sin 60°) = 43.3$ N m clockwise.

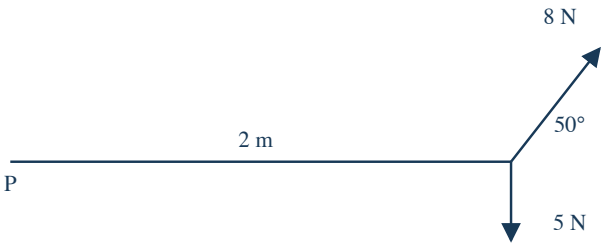
4–5

4. $10(2\sin 30°) = 10$ N m anticlockwise.
5. $5(3\cos 65°) = 6.34$ N m anticlockwise.

For a right triangle with hypotenuse $r$: opposite side $r\sin\theta$, adjacent side $r\cos\theta$.

## Test Your Understanding

Two forces act on a lamina. The 5 N force acts downwards, 2 m horizontally to the right of $P$. The 8 N force acts at 50° above a horizontal radius of length 2 m to the right of $P$. Find each moment.




**Solution 7**

5 N

$M_5 = 5(2) = 10\,\text{N m}$

clockwise.

8 N

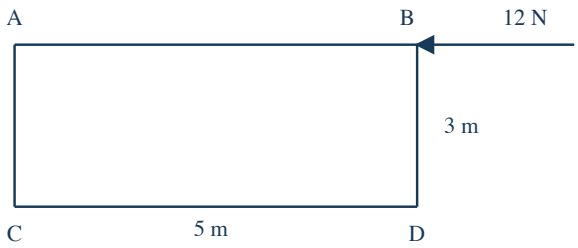
$M_8 = 8(2\sin 50°)$

$= 12.3\,\text{N m}$

anticlockwise (3 s.f.).

## Exercise 4A Page 72

4. A rectangle has $AB = CD = 5$ m and $AC = BD = 3$ m. A horizontal 12 N force acts to the left at $B$. Find its moment about a) $A$, b) $B$, c) $C$, d) $D$.




**Solution 8**

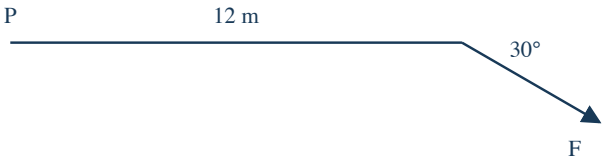
4a–d

a) 0. b) 0. Both points lie on the line of action.

c) $12(3) = 36\,\text{N m}$ anticlockwise.

d) $36\,\text{N m}$ anticlockwise.

5. A force $F$ acts at $30^\circ$ below a rod, 12 m to the right of pivot $P$. Its moment is 15 N m clockwise. Find $F$.




Solution 9

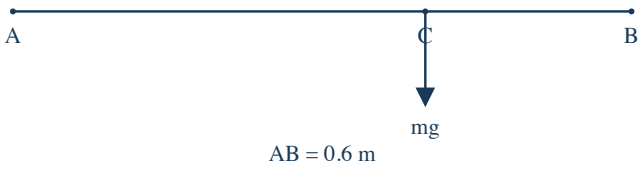
5

$15 = F(12 \sin 30^\circ) = 6F$

$F = 2.5$ N.

## Extension

A mass hangs at $C$ on a horizontal beam $AB$ of length 0.6 m. The magnitude of its moment about $A$ is twice that about $B$. Find $AC$.




Solution 10

Working

Let $AC = x$. Then $BC = 0.6 - x$.

$mgx = 2mg(0.6 - x)$

$3x = 1.2 \quad \Rightarrow \quad AC = 0.4$ m.

## Resultant moments

Choose clockwise or anticlockwise as positive and add signed moments. Find these resultants about $P$.

1. 20 N downwards, 10 m right; 10 N downwards, 15 m left.
2. 5 N downwards, 3 m right; 4 N upwards, 1 m right; 3 N downwards, 1 m left.
3. A light rod carries 50 kg at 1 m left of $P$, and 30 kg at 3 m right of $P$.
4. A light rod has length $x$ m. Pivot $P$ is 4 m from its left end; a 30 kg mass hangs at its right end ($x > 4$).

1–2

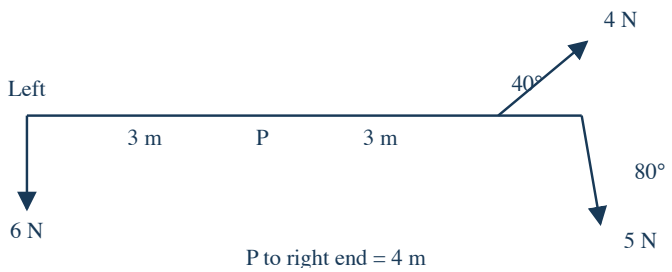
1. $200 - 150 = 50$ N m clockwise.
2. $15 - 4 - 3 = 8$ N m clockwise.

3–4

3. $30g(3) - 50g(1) = 40g$ N m clockwise.
4. $30g(x - 4)$ N m clockwise.



A light horizontal rod has pivot $P$. A 6 N downward force acts 3 m left of $P$; a 4 N force acts 3 m right, at 40° above the rod; a 5 N force acts 4 m right, at 80° below the rod. Find the resultant moment.




Distances

Perpendicular distances are 3, $3\sin 40°$, and $4\sin 80°$ metres.

Resultant

Taking anticlockwise as positive:

$M = 6(3) + 4(3\sin 40°)$

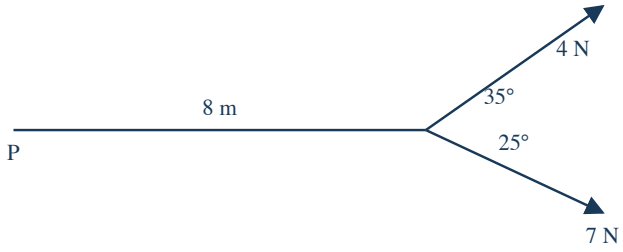
$-5(4\sin 80°)$

$= 6.02$ N m anticlockwise.

## Test Your Understanding

**Solution 13**

Two forces act at a point 8 m horizontally to the right of $P$ on a lamina: 7 N at 25° below the horizontal and 4 N at 35° above it. Find the resultant moment.




Distances

The perpendicular distances are $8\sin 25°$ and $8\sin 35°$ metres.

Resultant

Taking clockwise as positive:

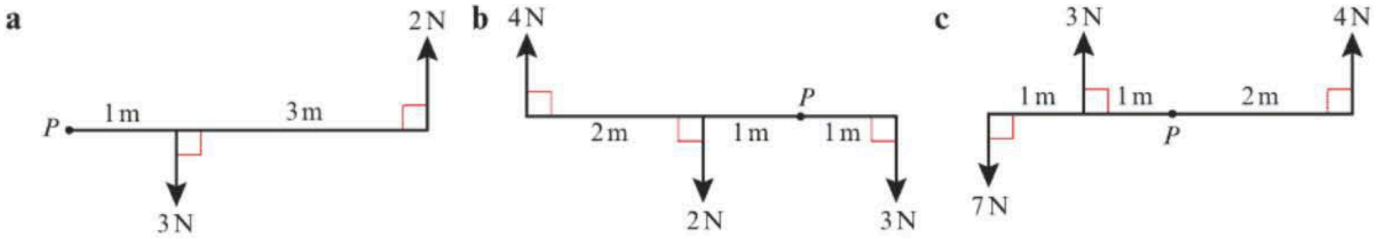
$M = 7(8\sin 25°) - 4(8\sin 35°)$

$= 5.31$ N m clockwise.

## Exercise 4B Page 74-76

**Solution 14**

1. Find the resultant moment about $P$ for each light rod.

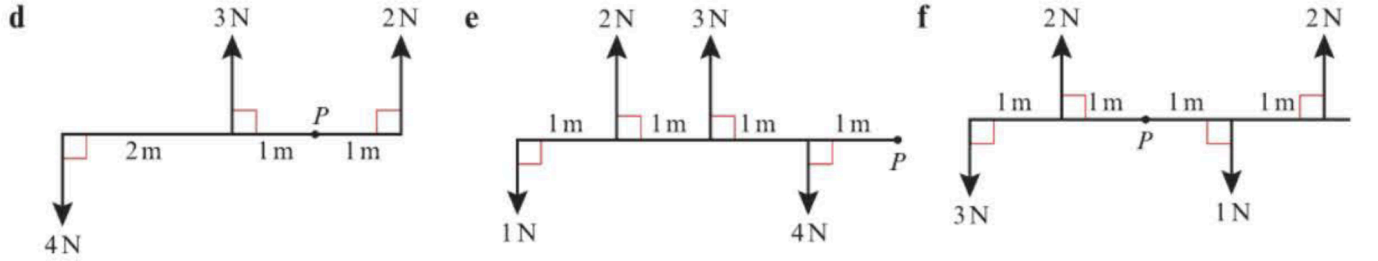



a–c

a) $8 - 3 = 5$ N m anticlockwise.

b) $12 - 2 + 3 = 13$ N m clockwise.

c) $14 - 3 + 8 = 19$ N m anticlockwise.



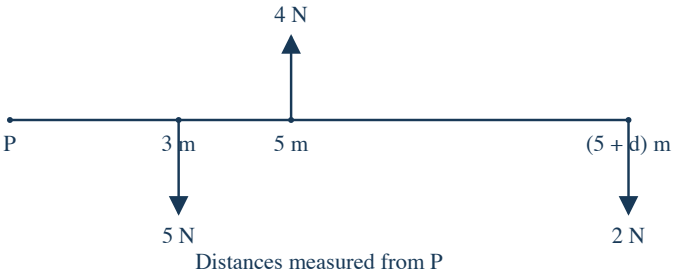

d–f

d) $12 - 3 + 2 = 11$ N m anticlockwise.

e) $-4 + 6 + 6 - 4 = 4$ N m clockwise.

f) $6 - 2 - 1 + 4 = 7$ N m anticlockwise.

3. Forces act on a light rod as shown. The resultant moment about $P$ is 17 N m clockwise. Find $d$.



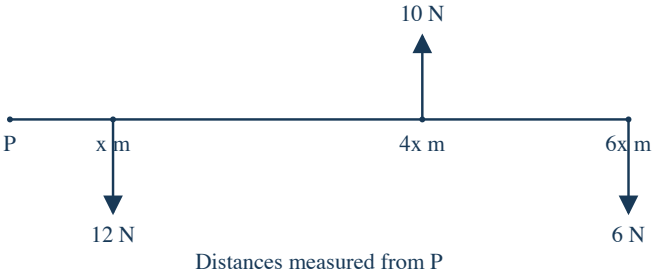

3

$2(5+d)+5(3)-4(5)=17$

$5+2d=17 \quad \Rightarrow \quad d=6 \text{ m.}$



4. The resultant moment about $P$ is 12.8 N m clockwise. Find $x$.

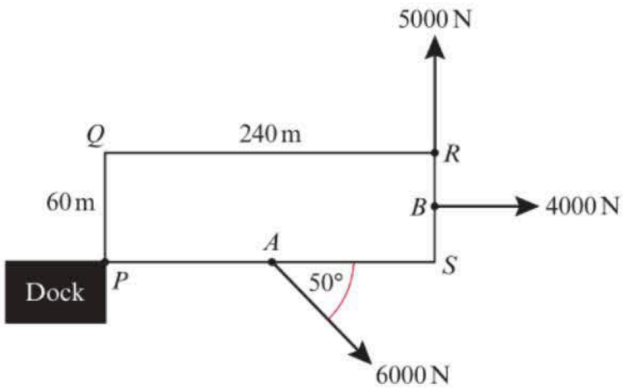



4

$6(6x)+12x-10(4x)=12.8$

$8x=12.8 \quad \Rightarrow \quad x=1.6 \text{ m.}$

5. A cruise ship is modelled as a rectangular lamina $PQRS$, fixed at $P$. Three tugs exert the forces shown. $A$ is the midpoint of $PS$, and $B$ the midpoint of $RS$. Find the resultant moment about $P$ and its direction.




5

Take anticlockwise as positive.

$M = 5000(240) - 4000(30)$

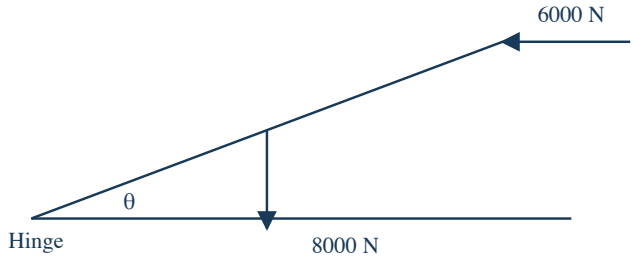
$-6000(120\sin 50°)$

$= 528448\,\text{N m}$

$= 528\,\text{kN m}$ anticlockwise (3 s.f.).



6. A uniform drawbridge of weight 8000 N is inclined at $\theta$ above the horizontal. A horizontal 6000 N force at its upper end acts towards the hinge to raise it. Show that raising requires $\tan\theta > 2/3$.




6

Let the bridge length be $L$. For raising:

$6000L\sin\theta > 8000\dfrac{L}{2}\cos\theta.$

$\tan\theta > \dfrac{2}{3} \quad (0 < \theta < 90°).$

## Extension

Solution 19

1. A rectangular barge is 120 m long and 20 m wide. It is fixed at its bottom-right corner $P$. At the top-left, a 2000 N force acts down-left at $45^\circ$ to the vertical; at the bottom-left, 1100 N acts down-right at $60^\circ$ below the horizontal. Find the resultant moment and state a modelling assumption.

1

Anticlockwise moment:

$2000(120\cos 45^\circ + 20\sin 45^\circ)$

$+1100(120\sin 60^\circ) = 332309\ \text{N m}.$

332 kN m anticlockwise. Model the barge as a rigid lamina, with no other turning forces.

2. A uniform aircraft door $AB$, length 4 m and weight 2500 N, is hinged at $B$. It is inclined at $45^\circ$ to the horizontal. A vertical upward force at $A$ tends to lift the door. Find the force needed to start lifting.

2

At the threshold:

$F(4\cos 45^\circ) = 2500(2\cos 45^\circ).$

$F = 1250$ N.

A force greater than this starts lifting.

## This whole chapter in a nutshell...

Solution 20

**Equilibrium of a rigid body**

Both conditions are required:

$$\sum \mathbf{F} = \mathbf{0}, \qquad \sum M = 0.$$

For vertical forces, total upward force equals total downward force. About any point, total clockwise moment equals total anticlockwise moment.

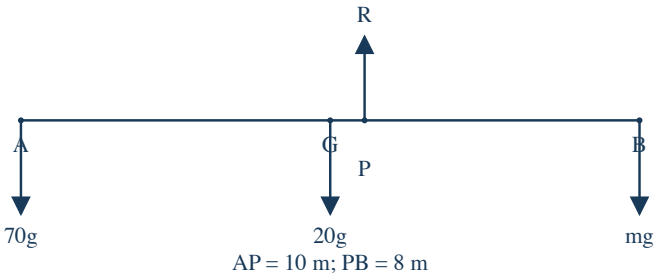
**A reliable method**

1. Draw the external forces.
2. Place a uniform rod's weight at its midpoint.
3. Resolve forces.
4. Take moments about a point that eliminates an unwanted force.
5. Check units and the physical meaning of the answer.

## Example

**Solution 21**

A uniform seesaw has length 18 m and mass 20 kg. Its pivot $P$ is 10 m from the left end. Woody (70 kg) sits at the left end; Buzz ($m$ kg) sits at the right. Find $m$ and the pivot reaction $R$.




Method 1

Moments about $P$:

$70g(10) + 20g(1) = mg(8)$.

$m = 90$ kg.

Vertical forces give $R = 180g = 1764$ N.

Method 2

Alternatively, moments about $B$:

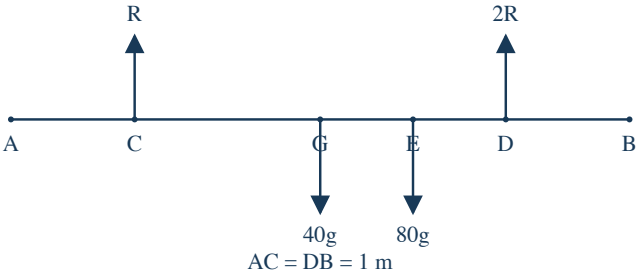
$8R = 70g(18) + 20g(9)$.

Hence $R = 1764$ N, then $R = (70 + 20 + m)g$ gives $m = 90$ kg.

## Two Pivots

**Solution 22**

A uniform beam $AB$, length 5 m and mass 40 kg, rests on supports $C, D$ with $AC = DB = 1$ m. An 80 kg man stands at $E$. The reaction at $D$ is twice that at $C$. Find $AE$.




Working

$3R = 120g \quad \Rightarrow \quad R = 40g$.

Let $AE = x$. Moments about $A$:

$40g(1) + 80g(4) = 40g(2.5) + 80gx$.

$x = 3.25$ m.

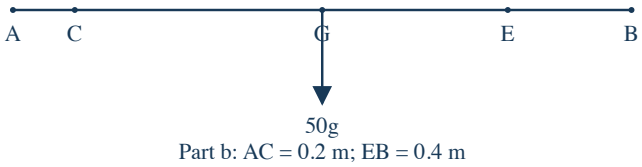
Include the beam's 40 kg mass; man–beam contact forces are internal to the combined system.

## Test Your Understanding

Edexcel M1 May 2013(R), Q8. A uniform beam $AB$ is 2 m long and has mass 50 kg. Supports are at $C, D$, where $AC = 0.2$ m and $DB = x$.

a) If $R_D = 2R_C$, find $x$.

b) Move $D$ to $E$, where $EB = 0.4$ m. Put a particle of mass $m$ at $B$. If $R_E = 4R_C$, find $m$.




50g

Part b: AC = 0.2 m; EB = 0.4 m

**Solution 23**

a

$3R_C = 50g$. Moments about $C$:

$2R_C(1.8 - x) = 50g(0.8)$.

$x = 0.6$ m.

b

Let $R_C = S$. Then $5S = (50 + m)g$.

Moments about $B$:

$50g(1) = S(1.8) + 4S(0.4) = 3.4S$.

$m = \dfrac{400}{17} = 23.5$ kg (3 s.f.).

## Exercise 4C Page 78-80

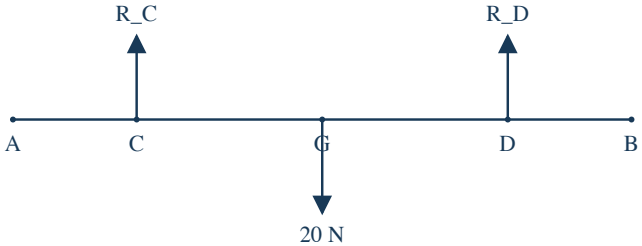
1. A uniform rod $AB$ is 5 m long and weighs 20 N. It rests horizontally at supports $C, D$. Find the reactions in each case. All distances are measured from $A$.

a) $AC = 1$ m, $AD = 4$ m.

b) $AC = 2$ m, $AD = 4$ m.

c) $AC = 1.5$ m, $AD = 4$ m.

d) $AC = 1.5$ m, $AD = 4.2$ m.




20 N

Schematic: support positions vary

**Solution 24**

a–d

In each case: $R_C + R_D = 20$. Take moments about $C$.

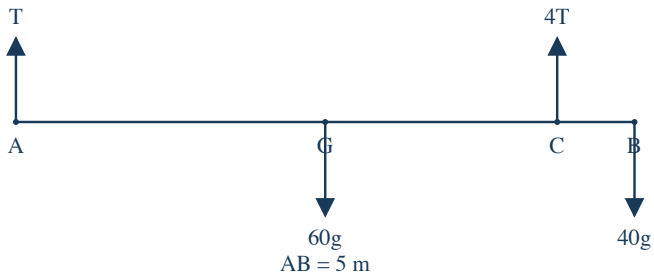
a) $3R_D = 20(1.5)$: $R_D = 10$, $R_C = 10$ N.

b) $2R_D = 20(0.5)$: $R_D = 5$, $R_C = 15$ N.

c) $2.5R_D = 20(1)$: $R_D = 8$, $R_C = 12$ N.

d) $2.7R_D = 20(1)$: $R_D = 7.41$, $R_C = 12.6$ N (3 s.f.).

7. A uniform beam $AB$, length 5 m and mass 60 kg, carries a 40 kg load at $B$. Vertical wires at $A, C$ hold it horizontally. The tension at $C$ is four times that at $A$. Find a) the tension at $C$, b) $CB$.




7a–b

a) $5T = 100g$, so $4T = 80g = 784$ N.

b) Let $CB = x$. Moments about $B$:

$20g(5) + 80gx = 60g(2.5)$.

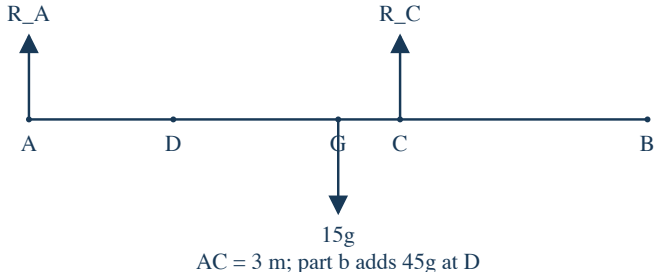
$x = 0.625$ m.



8. A uniform plank $AB$, length 5 m and mass 15 kg, rests at $A, C$, where $BC = 2$ m.

a) Find the reaction at $C$.

b) A 45 kg person stands at $D$, making the two reactions equal. Find $AD$.




8a–b

a) Moments about $A$:

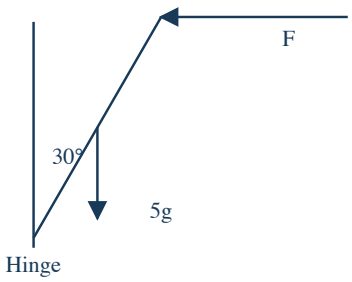
$3R_C = 15g(2.5)$, so $R_C = 122.5$ N.

b) $2R = 60g$, so $R = 30g$.

$45g(AD) + 15g(2.5) = 30g(3)$.

$AD = \frac{7}{6} = 1.17$ m (3 s.f.).

10. A uniform lever has mass 5 kg and length 1.5 m. It is smoothly hinged to a vertical wall and is held 30° from the vertical by a horizontal force $F$ at its other end. Find $F$.




10

Moments about the hinge:

$F(1.5\cos 30^\circ) = 5g(0.75\sin 30^\circ)$.

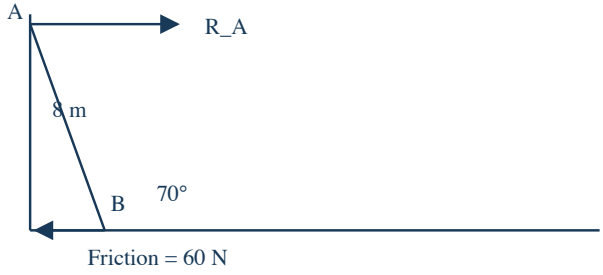
$F = 2.5g\tan 30^\circ = 14.1$ N (3 s.f.).



11. A uniform ladder $AB$, length 8 m, leans against a smooth vertical wall at 70° to rough horizontal ground. Friction at $B$ has magnitude 60 N.

a) Find the wall reaction at $A$.

b) Find the ladder's mass.




11a–b

a) Horizontal equilibrium gives $R_A = 60$ N.

b) Moments about $B$:

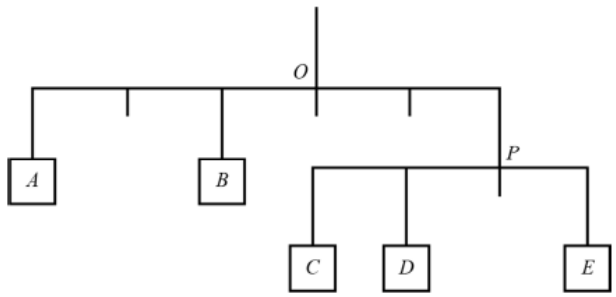
$60(8\sin 70^\circ) = mg(4\cos 70^\circ)$.

$m = \dfrac{120\tan 70^\circ}{g} = 33.6$ kg (3 s.f.).

## Extension

Solution 29

Arrange masses 1, 2, 3, 4 and 5 kg on the five marked positions of this kinetic sculpture. Use each once so both light rods hang horizontally. Marked intervals on each rod are equal.




Hints

For masses $A, B, C, D, E$, moments about the upper suspension give

$3A + B = 2(C + D + E)$.

Thus $A, B$ have the same parity. For the lower rod:

$2C + D = E$.

Arrangement

From left to right:

$3,\ 5,\ 1,\ 2,\ 4$ kg.

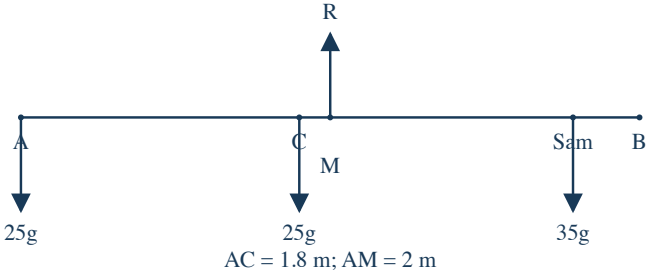
Check: $3(3) + 5 = 2(1 + 2 + 4) = 14$, and $2(1) + 2 = 4$.

## Centres of Mass

Solution 30

A uniform rod's centre of mass is its midpoint. For a non-uniform rod it need not be.

A non-uniform plank $AB$, length 4 m and mass 25 kg, is pivoted at its midpoint $M$. Its centre of mass $C$ has $AC = 1.8$ m. Tamsin (25 kg) sits at $A$. Where must Sam (35 kg) sit to balance it?




Working

Let Sam be $x$ metres from $A$. Moments about $M$:

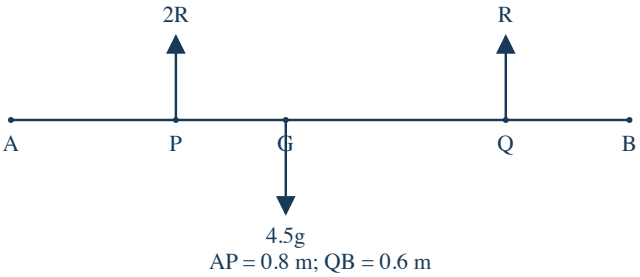
$25g(2) + 25g(0.2) = 35g(x - 2)$.

$x = \dfrac{25}{7} = 3.57$ m (3 s.f.).

## Test Your Understanding

**Solution 31**

Edexcel M1 May 2012, Q2. A non-uniform rod $AB$ has length 3 m and mass 4.5 kg. It rests at $P, Q$, with $AP = 0.8$ m and $QB = 0.6$ m. The reaction at $P$ is twice that at $Q$. Find a) the reaction at $Q$, b) the distance $AG$ to the centre of mass.




a–b

a) $3R = 4.5g$, so $R = 14.7$ N.

b) Moments about $A$:

$4.5g(AG) = 2R(0.8) + R(2.4)$.

$AG = \frac{4}{3}$ m.

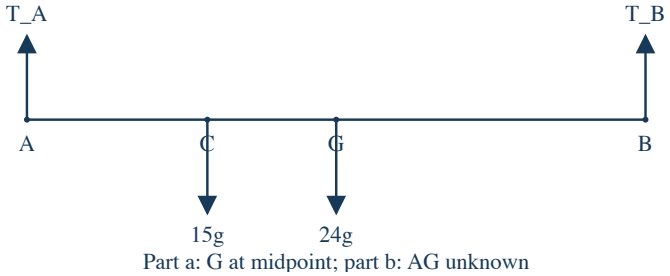
## Exercise 4D Page 81-83

**Solution 32**

5. A plank $AB$, mass 24 kg and length 4.8 m, carries a 15 kg load at $C$, where $AC = 1.4$ m. Vertical ropes at $A, B$ hold it horizontally.

a) Model the plank as uniform. Find the tension at $B$.

b) In a non-uniform model, the tension at $A$ is 25 N greater than at $B$. Find the distance of its centre of mass from $A$.




Part a: G at midpoint; part b: AG unknown

5a–b

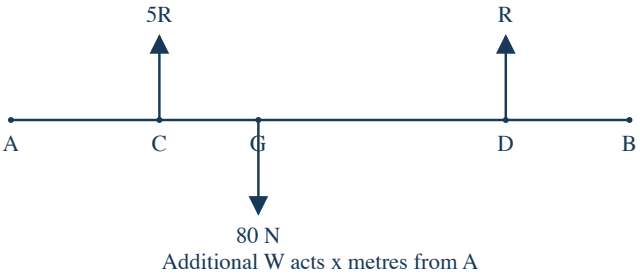
a) $4.8T_B = 15g(1.4) + 24g(2.4)$, so $T_B = 160.475 \approx 160$ N.

b) $2T_B + 25 = 39g$, giving $T_B = 178.6$ N.

$15g(1.4) + 24g(AG) = 178.6(4.8)$.

$AG = 2.77$ m (3 s.f.).

7. A non-uniform rod $AB$ has length 25 m and weight 80 N. Supports $C, D$ are 6 m and 20 m from $A$; its centre of mass is 10 m from $A$. A weight $W$ is attached $x$ metres from $A$. If $R_C = 5R_D$, show that $W = 400/(25 - 3x)$.




7

$6R = 80 + W.$

Moments about $A$:

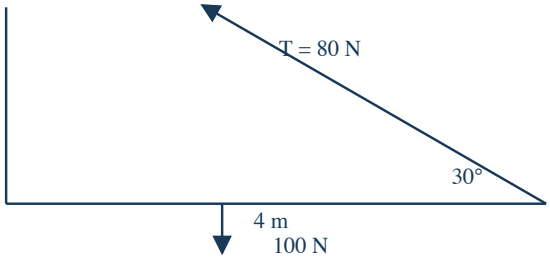
$5R(6) + R(20) = 80(10) + Wx.$

$50(80 + W) = 4800 + 6Wx.$

$(50 - 6x)W = 800 \quad \Rightarrow \quad W = \dfrac{400}{25 - 3x}.$



8. A non-uniform rod, length 4 m and weight 100 N, is hinged to a vertical wall. A cable at its free end makes $30^\circ$ with the horizontal rod and has tension 80 N. Find the distance of the centre of mass from the wall.



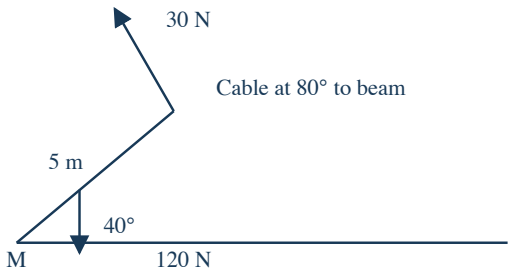

8

Moments about the hinge:

$100x = 80(4 \sin 30^\circ) = 160.$

$x = 1.6$ m.

Challenge. A non-uniform beam has weight 120 N and length 5 m. It is pivoted at $M$, inclined at 40° to the horizontal, and held by a cable at its upper end. Cable tension is 30 N at 80° to the beam. Find the distance of its centre of mass from $M$.




Solution 35

Working

Let the centre of mass be $x$ metres along the beam.

$120x\cos 40^\circ = 30(5\sin 80^\circ)$.

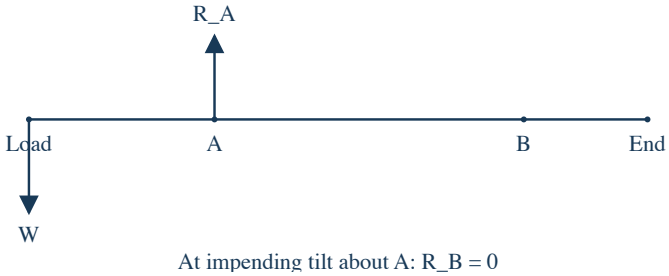
$x = \dfrac{5\sin 80^\circ}{4\cos 40^\circ} = 1.61$ m (3 s.f.).

## Tilting

### On the point of tilting

As a load moves beyond a support, the reaction at the other support decreases. At the instant contact is lost, that reaction is zero.

A cable that becomes slack likewise has zero tension.




At impending tilt about A: R_B = 0

Solution 36

Key principle

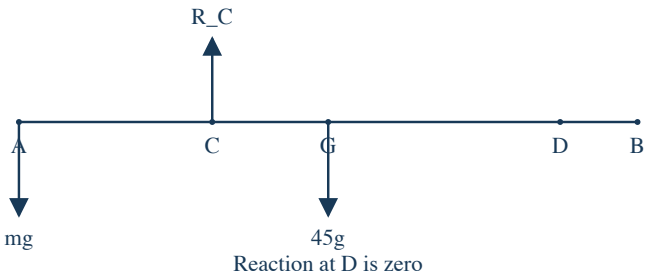
Draw the forces at the threshold and set the reaction at the support losing contact to zero. Take moments about the remaining pivot.

This condition applies to the support losing contact, not automatically to every other force or cable.

## Example

**Solution 37**

A uniform beam $AB$ has mass 45 kg and length 16 m. It rests at $C, D$, with $AC = 5$ m and $CD = 9$ m. When Woody stands at $A$, it is on the point of tilting about $C$. Find his mass.




Working

Moments about $C$:

$mg(5) = 45g(3)$.

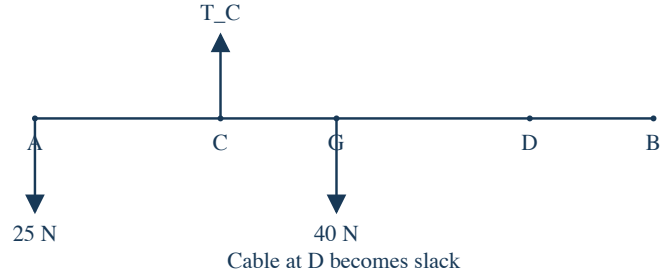
$m = 27$ kg.

Taking moments about $C$ eliminates its unknown reaction.

## Suspended System Example

**Solution 38**

A non-uniform rod $AB$, length 10 m and weight 40 N, hangs from cables at $C, D$, where $AC = 3$ m and $BD = 2$ m. When a 25 N weight is hung at $A$, it is on the point of rotating. Find its centre of mass's distance from $A$.




Working

The rod is about to turn about $C$, so $T_D = 0$. Let $AG = x$.

$25(3) = 40(x - 3)$.

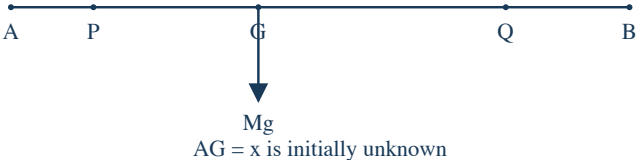
$x = 4.875$ m.

## Test Your Understanding

Edexcel M1 May 2013, Q6. A non-uniform beam $AB$, length 15 m, rests at $P, Q$, where $AP = 2$ m and $QB = 3$ m. A 50 kg child at $A$ brings it to the point of tilting about $P$; the same child at $B$ brings it to the point of tilting about $Q$.

a) Find the beam's mass and centre of mass.

b) The child stands at $X$, making the reactions equal. Find $AX$.




**Solution 39**

a

At the two thresholds:

$M(x-2) = 100, \qquad M(12-x) = 150.$

Add: $10M = 250$, so $M = 25$ kg. Then $x = 6$ m.

b

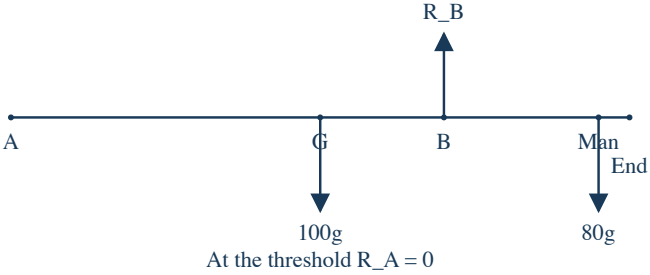
$2R = 75g$, so $R = 37.5g$. Moments about $A$:

$R(2+12) = 25g(6) + 50g(AX).$

$AX = 7.5$ m.

## Exercise 4E Page 84-85

5. A uniform plank has mass 100 kg and length 10 m. It rests on supports at its left end $A$ and at $B$, 7 m from $A$. An 80 kg man walks from $A$ towards the other end. How far can he walk past $B$ before tipping begins?




**Solution 40**

5

Let the man be $x$ metres beyond $B$.

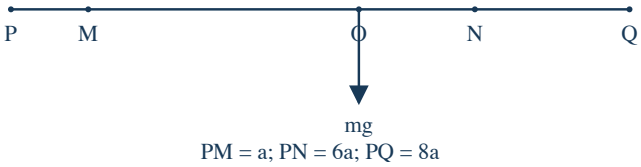
$100g(7-5) = 80gx.$

$x = 2.5$ m.

6. A non-uniform beam $PQ$, length $8a$ and mass $m$, hangs from wires at $M, N$, with $PM = a$, $QN = 2a$. A particle of mass $\frac{3}{4}m$ at $Q$ brings it to the point of tipping about $N$.

a) Show that $ON = \frac{3}{2}a$, where $O$ is the centre of mass.

b) Move the particle to the midpoint. Find the tension at $N$.



PM = a; PN = 6a; PQ = 8a

a

At tipping, $T_M = 0$. Moments about $N$:

$mg(ON) = \frac{3}{4}mg(2a)$.

$ON = \frac{3}{2}a$.

b

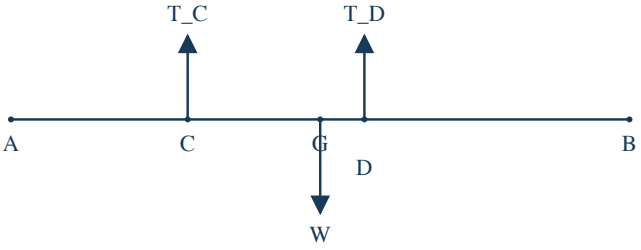
$PO = 6a - 1.5a = 4.5a$. Moments about $M$ after moving the particle:

$T_N(5a) = mg(3.5a) + \frac{3}{4}mg(3a)$.

$T_N = \frac{23}{20}mg$.



7. A uniform beam $AB$ has length 14 m and weight $W$. It hangs from vertical cables at $C, D$, with $AC = 4$ m and $BD = 6$ m. A 180 N weight at $A$ brings it to the point of tilting. Replace it with a weight $V$ at $B$. Find the maximum $V$ for equilibrium.



First add 180 N at A; then replace by V at B

7

First, $T_D = 0$. Moments about $C$:

$180(4) = 3W \quad \Rightarrow \quad W = 240$ N.

At maximum $V$, $T_C = 0$. About $D$:

$240(1) = 6V \quad \Rightarrow \quad V = 40$ N.